

TrongTin Tran Mai

Springfield, VA

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EDUCATION

George Mason University

Fairfax, VA

B.S in Computer Science; GPA: 3.87

Expected May 2028

- Relevant Coursework: Linear Algebra, Discrete Mathematics, Multivariable Calculus, Low-Level C Programming, Data Structures, Theory of Computation, Statistics for Engineers and Scientists

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, Swift, SQL, HTML/CSS

Frameworks & Tools: React.js, React Native, Node.js, FastAPI, PyQt5, Supabase, Railway, Git, Android Studio

Databases: PostgreSQL, SQLite

EXPERIENCE

Parkeye

Remote

Software Engineer

Sept 2025 – June 2026

- Developing a **cross-platform mobile application** for **500+ potential users**, using **real-time data** and location-aware logic to identify optimal parking based on **permit type, occupancy trends**, events, and campus conditions.
- Designed **PostgreSQL relational database schemas** in **Supabase**, implementing **authentication**, user preferences, parking data models, and **real-time synchronization** across **5+ database tables**.
- Developed and integrated **10+ REST API endpoints** and **backend services** to process parking availability data and deliver dynamic information to mobile clients.
- Built **scalable cloud infrastructure** using **Railway** and **Supabase**, supporting data across **20+ parking locations** and enabling future campus-wide deployment.

PROJECTS

KiteView | *React Native, TypeScript, Supabase, PDF.js*

Sept 2026 – Present

- Building a **cross-platform AI PDF reader** (macOS-first via **react-native-macos**) that turns static and scanned documents into an interactive reading experience with tap-to-explain annotations.
- Architecting a **Turborepo monorepo** with shared **packages/core**, **packages/ui**, and **packages/pdf-engine**, targeting native macOS now and iOS/Android later from one codebase.
- Implementing **phrase annotation** through **Supabase Edge Functions** that combine dictionary lookup and **LLM paraphrasing**, personalized by stored **reading preferences** (level, language, tone, domain).
- Designing **PostgreSQL schemas** with **Row Level Security** for auth, profiles, reading preferences, and custom instructions, keeping API keys server-side in Edge Function secrets.
- Developing **PDF form-field detection** and **autofill/export** flows (AcroForm + heuristic detection) so users can fill detected fields and export annotated/filled PDFs.

Holoura | *Full Stack Development, Machine Learning*

May 2026 – Present

- Building an **AI-powered purchasing agent** that continuously monitors collectible product availability across **10+ retail and marketplace sources**.
- Developing a **full-stack application** using **React Native**, **FastAPI**, **Supabase**, and **PostgreSQL**, with **12+ REST API endpoints** connecting frontend, backend, and database services.
- Engineering **data ingestion pipelines** that aggregate **15,000+ collectible products**, pricing information, availability data, and marketplace metadata through **third-party APIs**.
- Designing **machine learning models** for **deal detection**, **demand forecasting**, and **personalized recommendations** using historical pricing and product-demand signals.

AI Music Transcription System | *Python, PyQt5, Deep Learning, DSP*

Aug 2024 – May 2025

- Researched and implemented **machine learning** and **digital signal processing** techniques for automated audio-to-score transcription.
- Developed a **Python-based data processing pipeline** that converts WAV recordings into **sheet music notation** and PDF outputs, reducing manual transcription time by approximately **92%**.
- Evaluated **model performance** across **50+ test recordings**, achieving approximately **89% transcription accuracy** on tested material.